

Data Base Lab
Lab Task 8



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Question 01

```
MariaDB [lab_11_task]> START TRANSACTION;
Query OK, 0 rows affected (0.001 sec)

MariaDB [lab_11_task]> UPDATE accounts
  -> SET balance = balance - 1000
  -> WHERE account_id = 101;
Query OK, 1 row affected (0.004 sec)
Rows matched: 1 Changed: 1 Warnings: 0

MariaDB [lab_11_task]> UPDATE accounts
  -> SET balance = balance + 1000
  -> WHERE account_id = 102;
Query OK, 1 row affected (0.001 sec)
Rows matched: 1 Changed: 1 Warnings: 0

MariaDB [lab_11_task]> INSERT INTO transactions (from_account, to_account, amount, transaction_date, status)
  -> VALUES (101, 102, 1000, NOW(), 'Completed');
Query OK, 1 row affected (0.001 sec)

MariaDB [lab_11_task]> COMMIT;
Query OK, 0 rows affected (0.005 sec)

MariaDB [lab_11_task]> SELECT * FROM accounts WHERE account_id IN (101, 102);
+-----+-----+-----+-----+
| account_id | customer_id | balance | account_type |
+-----+-----+-----+-----+
|          101 |          1 | 4000.00 | Savings      |
|          102 |          2 | 4000.00 | Current      |
+-----+-----+-----+-----+
2 rows in set (0.001 sec)

MariaDB [lab_11_task]> SELECT * FROM transactions ORDER BY transaction_id DESC LIMIT 1;
+-----+-----+-----+-----+-----+-----+
| transaction_id | from_account | to_account | amount | transaction_date | status |
+-----+-----+-----+-----+-----+-----+
|              1 |          101 |          102 | 1000.00 | 2026-04-17 07:48:02 | Completed |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.001 sec)

MariaDB [lab_11_task]> |
```

Question 02

```
START TRANSACTION;
Query OK, 0 rows affected (0.001 sec)

MariaDB [lab_11_task]> SELECT account_id, balance FROM accounts WHERE account_id IN (102, 103);
+-----+-----+
| account_id | balance |
+-----+-----+
|          102 | 4000.00 |
|          103 | 7000.00 |
+-----+-----+
2 rows in set (0.001 sec)

MariaDB [lab_11_task]> UPDATE accounts
  -> SET balance = balance - 2000
  -> WHERE account_id = 102;
Query OK, 1 row affected (0.004 sec)
Rows matched: 1  Changed: 1  Warnings: 0

MariaDB [lab_11_task]> UPDATE accounts
  -> SET balance = balance + 2000
  -> WHERE account_id = 103;
Query OK, 1 row affected (0.001 sec)
Rows matched: 1  Changed: 1  Warnings: 0

MariaDB [lab_11_task]> INSERT INTO transactions (from_account, to_account, amount, transaction_date, status)
  -> VALUES (102, 103, 2000, NOW(), 'Failed');
Query OK, 1 row affected (0.001 sec)

MariaDB [lab_11_task]> ROLLBACK;
Query OK, 0 rows affected (0.005 sec)

MariaDB [lab_11_task]> SELECT account_id, balance FROM accounts WHERE account_id IN (102, 103);
+-----+-----+
| account_id | balance |
+-----+-----+
|          102 | 4000.00 |
|          103 | 7000.00 |
+-----+-----+
2 rows in set (0.001 sec)

MariaDB [lab_11_task]> SELECT * FROM transactions ORDER BY transaction_id DESC LIMIT 1;
+-----+-----+-----+-----+-----+-----+
| transaction_id | from_account | to_account | amount | transaction_date | status |
+-----+-----+-----+-----+-----+-----+
|          1 |          101 |          102 | 1000.00 | 2026-04-17 07:48:02 | Completed |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.001 sec)

MariaDB [lab_11_task]> |
```

Question 03

```
MariaDB [lab_11_task]> START TRANSACTION;
Query OK, 0 rows affected (0.001 sec)

MariaDB [lab_11_task]> UPDATE accounts SET balance = balance - 500 WHERE account_id = 101;
Query OK, 1 row affected (0.005 sec)
Rows matched: 1 Changed: 1 Warnings: 0

MariaDB [lab_11_task]> UPDATE accounts SET balance = balance + 500 WHERE account_id = 102;
Query OK, 1 row affected (0.001 sec)
Rows matched: 1 Changed: 1 Warnings: 0

MariaDB [lab_11_task]> INSERT INTO transactions (from_account, to_account, amount, transaction_date, status)
-> VALUES (101, 102, 500, NOW(), 'Completed');
Query OK, 1 row affected (0.001 sec)

MariaDB [lab_11_task]> SAVEPOINT after_first_transfer;
Query OK, 0 rows affected (0.000 sec)

MariaDB [lab_11_task]> UPDATE accounts SET balance = balance - 300 WHERE account_id = 102;
Query OK, 1 row affected (0.001 sec)
Rows matched: 1 Changed: 1 Warnings: 0

MariaDB [lab_11_task]> UPDATE accounts SET balance = balance + 300 WHERE account_id = 103;
Query OK, 1 row affected (0.001 sec)
Rows matched: 1 Changed: 1 Warnings: 0

MariaDB [lab_11_task]> INSERT INTO transactions (from_account, to_account, amount, transaction_date, status)
-> VALUES (102, 103, 300, NOW(), 'Completed');
Query OK, 1 row affected (0.001 sec)

MariaDB [lab_11_task]> SELECT 'Before rollback to savepoint' as Status;
+-----+
| Status |
+-----+
| Before rollback to savepoint |
+-----+
1 row in set (0.001 sec)

MariaDB [lab_11_task]> SELECT account_id, balance FROM accounts WHERE account_id IN (101, 102, 103);
+-----+-----+
| account_id | balance |
+-----+-----+
| 101 | 3500.00 |
| 102 | 4200.00 |
| 103 | 7300.00 |
+-----+-----+
3 rows in set (0.001 sec)

MariaDB [lab_11_task]> ROLLBACK TO SAVEPOINT after_first_transfer;
Query OK, 0 rows affected (0.001 sec)
```

```
MariaDB [lab_11_task]> ROLLBACK TO SAVEPOINT after_first_transfer;
Query OK, 0 rows affected (0.001 sec)

MariaDB [lab_11_task]> SELECT 'After rollback to savepoint' as Status;
+-----+
| Status |
+-----+
| After rollback to savepoint |
+-----+
1 row in set (0.000 sec)

MariaDB [lab_11_task]> SELECT account_id, balance FROM accounts WHERE account_id IN (101, 102, 103);
+-----+-----+
| account_id | balance |
+-----+-----+
| 101 | 3500.00 |
| 102 | 4500.00 |
| 103 | 7000.00 |
+-----+-----+
3 rows in set (0.001 sec)

MariaDB [lab_11_task]> COMMIT;
Query OK, 0 rows affected (0.004 sec)

MariaDB [lab_11_task]> SELECT 'Final state after commit' as Status;
+-----+
| Status |
+-----+
| Final state after commit |
+-----+
1 row in set (0.001 sec)

MariaDB [lab_11_task]> SELECT account_id, balance FROM accounts WHERE account_id IN (101, 102, 103);
+-----+-----+
| account_id | balance |
+-----+-----+
| 101 | 3500.00 |
| 102 | 4500.00 |
| 103 | 7000.00 |
+-----+-----+
3 rows in set (0.001 sec)

MariaDB [lab_11_task]> SELECT * FROM transactions ORDER BY transaction_id DESC LIMIT 3;
+-----+-----+-----+-----+-----+-----+
| transaction_id | from_account | to_account | amount | transaction_date | status |
+-----+-----+-----+-----+-----+-----+
| 3 | 101 | 102 | 500.00 | 2026-04-17 07:56:20 | Completed |
| 1 | 101 | 102 | 1000.00 | 2026-04-17 07:48:02 | Completed |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.001 sec)

MariaDB [lab_11_task]> |
```

Question 04

```
MariaDB [lab_11_task]> CREATE USER 'bank_clerk'@'localhost' IDENTIFIED BY 'clerk123';
Query OK, 0 rows affected (0.007 sec)

MariaDB [lab_11_task]> GRANT SELECT, INSERT ON lab_11_task.transaction_summary TO 'bank_clerk'@'localhost';
Query OK, 0 rows affected (0.007 sec)

MariaDB [lab_11_task]> SHOW GRANTS FOR 'bank_clerk'@'localhost';
+-----+
| Grants for bank_clerk@localhost |
+-----+
| GRANT USAGE ON *.* TO 'bank_clerk'@'localhost' IDENTIFIED BY PASSWORD '*14291630CAA49EA0417C5F1791D7B2B1B20CAD64' |
| GRANT SELECT, INSERT ON `lab_11_task`.`transaction_summary` TO 'bank_clerk'@'localhost' |
+-----+
2 rows in set (0.000 sec)

MariaDB [lab_11_task]> |
```

QUESTION 06

```
MariaDB [lab_11_task]> REVOKE INSERT ON lab_11_task.transaction_summary FROM 'bank_clerk'@'localhost';
Query OK, 0 rows affected (0.005 sec)

MariaDB [lab_11_task]> SHOW GRANTS FOR 'bank_clerk'@'localhost';
+-----+
| Grants for bank_clerk@localhost |
+-----+
| GRANT USAGE ON *.* TO 'bank_clerk'@'localhost' IDENTIFIED BY PASSWORD '*14291630CAA49EA0417C5F1791D7B2B1B20CAD64' |
| GRANT SELECT ON `lab_11_task`.`transaction_summary` TO 'bank_clerk'@'localhost' |
+-----+
2 rows in set (0.000 sec)

MariaDB [lab_11_task]> |
```

Question 05

```
MariaDB [lab_11_task]> CREATE USER 'bank_manager'@'localhost' IDENTIFIED BY 'manager123';
Query OK, 0 rows affected (0.030 sec)
```

```
MariaDB [lab_11_task]> GRANT SELECT, INSERT, UPDATE, DELETE, CREATE VIEW
-> ON lab_11_task.*
-> TO 'bank_manager'@'localhost';
Query OK, 0 rows affected (0.005 sec)
```

```
MariaDB [lab_11_task]> GRANT CREATE, DROP, ALTER, INDEX
-> ON lab_11_task.*
-> TO 'bank_manager'@'localhost';
Query OK, 0 rows affected (0.008 sec)
```

```
MariaDB [lab_11_task]> SHOW GRANTS FOR 'bank_manager'@'localhost';
+-----+
| Grants for bank_manager@localhost |
+-----+
| GRANT USAGE ON *.* TO 'bank_manager'@'localhost' IDENTIFIED BY PASSWORD '*1B2333B70420F3DB5F4F164A9B89E21810F06840' |
| GRANT SELECT, INSERT, UPDATE, DELETE, CREATE, DROP, INDEX, ALTER, CREATE VIEW ON 'lab_11_task'.* TO 'bank_manager'@'localhost' |
+-----+
2 rows in set (0.001 sec)
```

```
MariaDB [lab_11_task]> SELECT * FROM mysql.user WHERE User = 'bank_manager'\G
***** 1. row *****
      Host: localhost
      User: bank_manager
      Password: *1B2333B70420F3DB5F4F164A9B89E21810F06840
      Select_priv: N
      Insert_priv: N
      Update_priv: N
      Delete_priv: N
      Create_priv: N
      Drop_priv: N
      Reload_priv: N
      Shutdown_priv: N
      Process_priv: N
      File_priv: N
      Grant_priv: N
      References_priv: N
      Index_priv: N
      Alter_priv: N
      Show_db_priv: N
      Super_priv: N
      Create_tmp_table_priv: N
      Lock_tables_priv: N
      Execute_priv: N
      Repl_slave_priv: N
      Repl_client_priv: N
      Create_view_priv: N
      Show_view_priv: N
      Create_routine_priv: N
      Alter_routine_priv: N
      Create_user_priv: N
      Event_priv: N
      Trigger_priv: N
      Create_tablespace_priv: N
      Delete_history_priv: N
      ssl_type:
      ssl_cipher:
      x509_issuer:
      x509_subject:
      max_questions: 0
      max_updates: 0
      max_connections: 0
      max_user_connections: 0
      plugin: mysql_native_password
      authentication_string: *1B2333B70420F3DB5F4F164A9B89E21810F06840
      password_expired: N
      is_role: N
      default_role:
      max_statement_time: 0.000000
1 row in set (0.025 sec)

MariaDB [lab_11_task]> |
```

